Thesis title: Benchmarking for Anime Character Generation: A Compact, Reproducible Pipeline for Identity, Style, and Short-Sequence Coherence

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1 Background

Generative image models (diffusion, GANs) produce high-quality anime imagery, but there is no compact, reproducible benchmark that jointly evaluates character identity consistency, style fidelity, short-sequence temporal coherence, and background/POI plausibility for tourism-oriented anime content. This project builds a minimal, reproducible pipeline that (1) computes automated metrics (CLIP, LPIPS, MS-SSIM, identity cosine, SD/CV/PSS/OPG), (2) compares generated frames to a small reference gallery of real POI photos where available, and (3) validates automated signals against a small human pilot. The original contribution is a focused, deliverable benchmark and analysis workflow tailored to tourism-anime use cases (identity + background plausibility + short-sequence coherence).

2 Aims and objectives (purpose & main hypotheses)

Primary aim: produce a compact, reproducible benchmark and pilot evaluation for anime character generation in tourism scenes.
Hypotheses H1: A composite of automated metrics (CLIP cosine + LPIPS + MS-SSIM + stability scores) correlates with human judgments of identity/style/background plausibility (Spearman $\rho > 0.4$ in pilot). H2: Using a small reference gallery of real POI photos (pseudo-GT) improves automatic detection of background plausibility vs purely unsupervised checks. H3: Prompt sensitivity metrics (PSS, OPG) explain a meaningful portion of cross-role variability in identity consistency.

Objectives

- O1 Data: curate pilot dataset (200–300 images; ~6–8 characters) mixing public, synthetic, and collaborator-provided POI photos (reference gallery).
- O2 Metrics: implement Identity Consistency (embedding cosine), Style Fidelity (CLIP similarity, LPIPS), Structural Stability (MS-SSIM), Diversity (intra-class variance), and stability/prompt metrics (SD, CV, PSS, OPG).
- O3 Baselines: evaluate Stable Diffusion (base) and one public anime fine-tuned checkpoint; LoRA/rerank optional.
- O4 Human pilot: 3–5 raters; ~15–25 randomized pairwise items each; primary analyses: mean, SD, Spearman ρ, bootstrap CIs.
- O5 Reproducibility: publish code, prompts, seeds, metadata, embeddings; do not publish copyrighted raw images.

3 Research Design (data required; major methods / procedures)

Data required: pilot images (100–200) grouped by sequence/role and scene; small reference gallery of real POI photos (from collaborator Project1 or public sources) for background plausibility checks; generation metadata (prompt, seed, model, sampler, steps, guidance_scale).

Major methods / procedures

- Preprocessing: resize to 512×512, deduplicate (pHash), store metadata.
- Feature extraction: CLIP image embeddings; LPIPS and MS-SSIM pairwise computations.
- Automated checks: adjacent-frame and centroid comparisons (CLIP cosine, LPIPS, MS-SSIM); compute SD/CV/PSS/OPG per prompt/role. For background plausibility use CLIP(text→image) and CLIP(image→image) vs reference gallery; quantify perceptual gap via LPIPS/MS-SSIM to best reference.
- Anomaly detection & flagging: composite rule or simple anomaly model (isolation forest) to prioritize human review.
- Human validation: blind pairwise Likert ratings on identity, style, background plausibility, temporal continuity; use Spearman ρ and simple regression to map metrics → human scores; bootstrap CIs for uncertainty.
- Deliverables: reproducible pipeline (scripts), metadata.csv/per_sample.csv, thresholds and calibration notes, pilot

analysis report.

4 Ethical approvals

Human rater study requires IRB/ethics approval. A small-pilot application will be prepared immediately (3–5 raters, anonymized data, voucher compensation). Real POI photos used as internal reference only; copyrighted originals will not be published—only hashes, embeddings, and derived artifacts.

5 Resources (required & availability)

- Compute: UoA ML GPU server (RTX 3090 / A100 / H100) or cloud credits; storage ≥1 TB. Pilot generation target ~100 generated images; expand if resources permit. (Access request pending.)
- Software: Python, PyTorch, Hugging Face diffusers, CLIP, Faiss, FiftyOne, lpips, pytorch-msssim, statsmodels/scipy.
- Human resources: 3–5 raters from university participant pool; collaborators supplying POI photos and image sets.

6 Potential problems & alternative plans

- Compute limits: LoRA / heavy fine-tuning may be infeasible — treat as optional; reduce sample counts or delegate to lab server.
- Low rater power: pilot has limited power; accept wider CIs and use automated metrics for larger-scale signals; report Krippendorff's α and paired tests only if sample size supports them.
- Clustering/identity automation instability: DBSCAN/automatic clustering may be noisy for stylized anime — fallback to manual role grouping for core analyses.
- Metric–human mismatch: fallback to fine-tune an anime-specific embedding or build a composite regression predictor combining CLIP, LPIPS, MS-SSIM, and stability scores.

7 Tentative timeline (condensed)

- Wk3–4: submit proposal; ethics draft; repo skeleton; seminar PPT.
- Wk5: data selection (200–300); preprocessing; CLIP extraction; metadata schema.
- Wk6: run baselines; generate pilot (~100 generated images).
- Wk7 (early): compute automated metrics (CLIP, LPIPS, MS-SSIM, intraclass variance); initial SD/CV/PSS/OPG analysis.
- Wk7 (mid): human pilot (3–5 raters); collect ratings.
- Wk8: integrate automated + human results; finalize statistics; prepare seminar.
- Wk9: deliver seminar; finalize reproducible artifacts.
- Wk10–12: debugging, writing, thesis finalization and submission.

8 References

Borji, A. (2021). Pros and cons of GAN evaluation metrics: A comparative study. Computer Vision and Image Understanding.

Gehrmann, S., Strobelt, H., & Rush, A. M. (2019). GLTR: Statistical detection and human evaluation of generated text. Proceedings of ACL.

Zhang, R., Isola, P., Efros, A. A., Shechtman, E., & Wang, O. (2018). The unreasonable effectiveness of deep features as a perceptual metric. Proceedings of CVPR.

Yuan, H., et al. (2023). A massive multi-discipline multimodal understanding and reasoning benchmark for expert AGI. arXiv:2311.16502.

Supervisor's (Jiamou Liu has already verbally and messaged me that my proposal is okay to go ahead for now, can provide confirmation from him. Apologies that didn't obtain his signature right now)